


# ECO432

## Money and Banking

## "Antecedents and Consequences of the default risk of banks"

**Course Instructor: Professor Syed Abdulla Al Mamun**

## Term Paper

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## Introduction

The banking sector is very important for a country's financial stability. It also affects economic growth. Banks help by gathering savings and making investments. They act as middlemen in how capital is distributed. When banks face high default risk, they may struggle to meet their financial obligations. This includes paying back depositors and other creditors. This can harm their important role in the economy. Bank defaults can disrupt financial intermediation. The 2008 global financial crisis shows they can also cause big systemic issues. To keep a strong financial system, it is important to understand why default risk happens in banks and what it leads to.

Bangladesh depends on the strength of the banking sector to keep its steady economic development since it has a developing financial system that is more and more linked with worldwide capital markets. Apart from significantly helping to increase financial inclusion, banks in Bangladesh are the main means of loan distribution and deposit collecting in areas regarded as underdeveloped. Still, elements including asset quality, capital sufficiency, and deteriorating profitability regularly threaten this important company and raise the possibility of financial institutions failing in their responsibilities. Having a strong awareness of the elements of default risk and the wider consequences of those elements helps one to manage risks successfully, implement legislative changes, and build investor trust.

Four notable private commercial banks in Bangladesh—**BRAC Bank, Bank Asia, AB Bank, and Al-Arafah Islami Bank**—form the main emphasis of this study. This study looks at every one of these banks since each of them has a unique operating strategy, market concentration, and financial profile.

Default risk in financial organizations can arise from a mix of internal inefficiencies and outside economic shocks, most of which combine. Among the internal stakeholders of the company, poor financial performance, inadequate capital buffers, and poor asset quality all greatly increase the default risk. From the outside, macroeconomic events such as inflation, devaluation of currencies, and interest rate volatility can negatively affect the balance sheet quality of banks. Under this framework, legislators, investors, and regulatory authorities routinely check a range of financial indicators to project the probability of banks defaulting on their loans and to act preventatively in line with this projection.

The aim of this study is to identify the factors influencing bank default risk. Another name for these components is the antecedents, or underlying causes of default risk. Analyzing important financial metrics such as Capital Adequacy Ratio (CAR), Return on Assets (ROA), and Return on Equity (ROE) will help one to achieve this. Standard practice in the banking literature is the use of these indicators in order to evaluate the dependability and efficiency of financial organizations. These metrics give companies information on financial soundness, efficiency, and profitability. A higher ROA or ROE shows that the bank is able to perform both of these things at the same time, as a higher CAR implies that a bank is able to correctly manage its resources and deliver better returns to its shareholders. Conversely, a higher CAR suggests that the bank has enough capital to resist losses not expected from the start.

To evaluate the risk of insolvency, we apply the widely known Z-score, which considers both profitability and leverage and takes into account This enables us to examine the default risk and ascertain whether or not default is likely to arise. Higher Z-scores indicate that the default risk is less than it would otherwise be. We find the general stability of the banks by means of the Z-score, which considers several aspects of the financial situation of a bank.

Second main focus of the research is the consequences of default risk. More especially, the main focus of study is the extent to which the risk level influences market opinions and bank

valuation. We employ Tobin's Q (TQ), a market-based indicator of the ratio of a company's market value to the cost of replacing its assets, to achieve this. The TQ reflects the feelings of investors as well as their expectations about a bank's future performance. It is expected that a higher default risk would lower investor confidence, which will consequently affect Tobin's Q. Moreover, the ratio of non-performing loans, sometimes referred to as NPLs, is considered to be a control variable since poor asset quality directly affects the risk of default as well as investor impression.

Using two econometric models—one of which looks at the variables influencing the Z-score and the other of which links the Z-score to non-performing loans (NPLs) and Tobin's Q—this paper attempts to offer a complete view of the dynamics surrounding bank failure risk. The data that was used is panel data gathered from a range of commercial banks over multiple years in order to provide insights with both cross-sectional and time-series characteristics.

By offering empirical data on the following subjects:

- Which financial indicators are the most important drivers of the risk of default in banks, and
- How does such risk, once it has presented itself, influence market-based outcomes such as investor valuation?

By doing this, the study has helpful consequences for many stakeholders, including authorities, investors, and bank management, especially in developing countries where the financial industry is still prone to volatility. The results could also help to guide the development of laws aimed at enhancing surveillance of financial institutions, lowering systematic risk, and so boosting openness.

## Literature Review

The concept of default risk in banks has been thoroughly explored in the realms of finance and economics due to its implications for the stability of the financial system, market confidence, and regulatory oversight policies.

Kassim & Rahman (2009) provides an extensive empirical analysis of the approaches utilized by microfinance institutions (MFIs) in Bangladesh, including Grameen Bank, BRAC, and ASA, to implement effective risk management strategies aimed at efficiently mitigating default risk. The research was carried out in Bangladesh. This analysis examines the diverse strategies utilized by microfinance institutions (MFIs), including peer monitoring, adjustable repayment periods, dynamic incentives, and rigorous borrower screening as key approaches in their operational framework. The study indicates that microfinance institutions (MFIs) demonstrate resilience in the face of external shocks, exemplified by the floods of 1998. The long-term sustainability of microfinance institutions (MFIs) can be linked to the active engagement of their clientele and the flexibility of their repayment options. Moreover, it underscores the importance of institutional discipline and robust organizational frameworks as factors that play a crucial role in reducing default rates through their engagement. This outcome is realized through the input of these components. The research findings suggest that the sustainability of microfinance in Bangladesh can be attributed to a confluence of social cohesion, institutional innovation, and robust financial governance. This conclusion is derived from the empirical findings of the research conducted. This highlights the effectiveness of tailored risk management solutions within the microfinance sector globally, which ultimately provides essential insights.

Chowdhury and Dhar (2012) examined the difficulties encountered by Bangladesh's commercial banking sector as a result of loan defaults in great detail. A study was carried out to compare the efficiency of private commercial banks with that of state-owned commercial banks. We were able to identify the main variables contributing to these issues and the main causes that have led to them. Over the course of the study's time frame, PCBs consistently outperformed SCBs in terms of credit performance, loan recovery rates, and compliance with corporate governance requirements. Lack of transparency was a major factor in the increase of non-performing loans (NPLs). Political involvement, ineffective credit management, weak company governance, and ineffective legal enforcement were among the factors that contributed to this accumulation. Many factors make SCBs vulnerable to risk. These include political meddling in loan distribution, weak internal audit systems, and not adhering to required capital adequacy ratios. These issues do, in fact, exist, as has been demonstrated. Despite numerous reform programs, such as the Financial Sector Reform Program (FSRP) and compliance with Basel II criteria, South-Central Banks (SCBs) continue to have substantial difficulties pertaining to non-performing loans (NPLs) and governance concerns. Because of this, South-Central Banks (SCBs) face a formidable obstacle. In order to stabilize the banking industry and solve the problem of defaults, the authors propose the following measures: increasing the independence of the Bangladesh Bank; improving governance structures; lowering political influence; and increasing legal enforcement.

Hossain (2018) claims that non-performing loans (NPLs) have become rather common in Bangladesh's banking system. These loans, in his view, seriously hinder the preservation of financial stability as well as economic growth. According to the poll results, the number of loans falling into default practically doubled between 2010 and 2017 during those two years. Root causes of this increasing trend were several structural problems, including inadequate governance, inadequate legal enforcement, and politically driven loans. While private banks

and banks connected to other countries have extraordinary credit management, banks controlled by the government and the state have shown to be disproportionately affected. Hossain claims that inaccurate borrower appraisal, political meddling, poor capital provisions, and insufficient recovery strategies have all helped to aggravate the crisis. Support for his argument comes from both national statistics and foreign comparisons. Among the steps recommended in this paper as possible remedies for the credit risk in emerging nations like Bangladesh are more regulatory inspection, Basel III implementation, and depoliticizing of bank governance. These developments offer interesting fresh angles.

Rahim and Nasreen (2022) conducted one of the research studies on the general effects on Bangladesh's financial and economic systems of default loans, more specifically non-performing loans (NPLs). The researchers found important elements that might be utilized to forecast non-performing loans (NPLs) by applying time-series analysis and a basic econometric model. Among these elements were poor borrower income, massive credit amounts, and outrageous lending rates. The results of the survey revealed that both state-owned and commercial banks are finding non-performing loans, sometimes referred to as NPLs, to be a growing, expensive problem. The research indicates that the main causes of this problem are inadequate political and governance motivation as well as politically driven loans. Between 2010 and 2018, non-performing loans—also known as NPLs—rose from 7.3% to 10.4%, with a lot of institutions, most notably Janata Bank, seeing notable rises in the defaulted loan count. Non-performing loans, sometimes referred to as NPLs, have broad effects. Among these effects are a loss of capital, crises concerning liquidity, less efficient banking, and finally a slowing down of economic growth. The writers voiced their displeasure with too liberal regulatory policies, including laws forgiving significant amounts of debt with fines of only a few dollars. Among the things they recommended as a means of addressing the problem were a package of activities covering more severe loan screening,

more robust legal enforcement, dependable collateral systems, reward systems for timely payback, and a culture change toward ethical borrowing. The results of their study indicate that to guarantee that Bangladesh's financial situation is entirely stable, governance, enforcement, and institutional responsibility must be strengthened.

Akter and Roy (2017) conducted empirical research to ascertain how non-performing loans (NPLs) affect the profitability of commercial banks listed on the Dhaka Stock Exchange (DSE) between the years 2008 and 2013. This study effort aimed to investigate the link among four fundamental components and the net profit margin (NPM). Other elements were the interest margin, also known as IM; the loan-to-deposit ratio, sometimes known as LDR; poor debt, sometimes known as BD; and the ratio of category loans to total loans, thereby indicating CLT. It was judged that the method of regression analysis was suitable for the analysis. Larger degrees of non-performing loans (NPLs) clearly have a detrimental effect on financial institution profitability. This is true because NPLs exhibit a statistically significant inverse link with profitability. The results unequivocally show that this link does in fact exist. The writers have underlined a range of issues as the main causes of the rising trend of non-performance of loans in Bangladesh. Contributing elements aggravating this problem are insufficient loan monitoring, credit risk management, and debt payback policies for collateral. They advised doing a thorough credit analysis, following more strict documentation, and staying to the guidelines developed by the central bank in order to lower the number of non-performing loans (NPLs) and preserve the stability of the banking system. Following advice created from the motivation helps one to achieve the previously mentioned aims. Based on the unambiguous results of this study, it is imperative to implement adjustments and enhance enforcement to prevent a decline in the financial performance of the sector. One can reach this conclusion based on the results of this investigation that is reasonable.

## Methodology

This study has used secondary data extracted from the annual financial statements (Consolidated) of Brac Bank PLC, AB Bank PLC, Al-Arafah Islami Bank PLC, and Bank Asia PLC to examine the banks' overall financial performances over 11 years (2013-2023).

| Name of the bank | Year | Total Loan & Advances | Total Assets (TA) | Total Deposit | Total Liabilities | Book Value of Equity (TE) | Market Value of Equity (Market Cap) | Interest Income | Earnings before Interest and Taxes (EBIT) | Retained Earnings (RE) |
|---|---|---|---|---|---|---|---|---|---|---|
| BRAC BANK PLC | 2013 | 119,514,547,936 | 185,576,398,523 | 124,746,344,037 | 172,552,343,579 | 13,024,054,944 | 4,579,336,270 | 18,715,692,063 | 6,141,204,313 | 5,171,126,214 |
| | 2014 | 124,299,994,135 | 209,733,601,225 | 146,366,349,410 | 190,445,025,450 | 19,288,575,775 | 10,660,588,435 | 22,269,682,075 | 7,282,663,641 | 6,034,620,214 |
| | 2015 | 149,934,139,696 | 231,822,841,330 | 142,648,188,939 | 211,629,590,121 | 20,193,251,209 | 13,958,774,477 | 17,700,287,627 | 8,079,577,777 | 4,088,724,423 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2016 | 175,841,420,944 | 268,324,018,065 | 181,478,777,560 | 246,059,911,627 | 22,264,106,438 | 18,400,315,969 | 18,310,392,086 | 9,851,481,392 | 5,524,376,341 |
| | 2017 | 203,431,019,401 | 319,550,216,302 | 216,929,919,763 | 292,950,072,688 | 26,600,143,614 | 44,949,821,517 | 21,734,412,978 | 9,436,773,363 | 8,601,051,187 |
| | 2018 | 238,400,399,660 | 358,004,626,083 | 255,073,828,613 | 322,754,563,360 | 35,250,062,723 | 47,254,362,557 | 27,463,966,939 | 9,395,373,850 | 13,342,589,185 |
| | 2019 | 264,870,263,857 | 414,855,096,444 | 297,754,929,671 | 374,272,689,086 | 40,582,407,358 | 49,088,338,015 | 32,622,685,991 | 8,961,551,953 | 15,230,405,214 |
| | 2020 | 273,438,940,961 | 456,595,103,748 | 333,615,619,705 | 408,027,138,316 | 48,567,965,432 | 44,005,906,618 | 27,079,940,016 | 8,193,761,025 | 16,346,550,011 |
| | 2021 | 322,135,188,439 | 527,912,923,582 | 364,337,612,226 | 450,964,833,084 | 76,948,090,498 | 60,559,499,357 | 24,225,217,948 | 10,128,399,417 | 26,402,962,532 |
| | 2022 | 412,084,750,584 | 655,937,320,087 | 447,058,816,329 | 574,907,709,477 | 81,029,610,610 | 48,728,818,280 | 31,365,476,735 | 11,451,467,138 | 29,086,051,630 |
| | 2023 | 520,287,630,605 | 837,145,184,672 | 588,430,345,153 | 750,124,781,520 | 87,020,403,152 | 52,367,391,339 | 41,863,962,190 | 15,745,228,715 | 33,197,686,658 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **AB BANK PLC** | 2013 | 140,121,297,505 | 208,005,543,738 | 161,846,275,315 | 191,065,494,672 | 16,940,049,066 | 7,245,425,475 | 17,953,048,673 | 5,289,658,349 | 5,171,126,214 |
| | 2014 | 177,570,797,390 | 254,668,156,050 | 198,189,197,221 | 236,511,042,476 | 18,157,113,574 | 9,461,799,131 | 21,398,789,847 | 6,924,844,673 | 6,034,620,214 |
| | 2015 | 209,725,203,267 | 285,010,254,719 | 213,818,905,499 | 262,218,029,374 | 22,792,225,345 | 8,374,251,313 | 20,811,576,016 | 5,020,190,334 | 7,435,241,593 |
| | 2016 | 218,769,451,248 | 314,835,891,011 | 245,640,782,909 | 291,721,430,830 | 23,114,460,181 | 11,307,934,843 | 18,914,027,627 | 5,019,435,593 | 7,836,238,115 |
| | 2017 | 229,647,286,231 | 314,564,999,466 | 235,954,308,453 | 291,793,754,377 | 22,771,245,089 | 14,184,618,19 | 17,566,941,274 | 4,476,122,637 | 7,051,767,695 |
| | 2018 | 241,070,141,407 | 322,525,971,619 | 235,444,849,491 | 299,875,489,746 | 22,650,481,873 | 7,702,604,000 | 19,467,459,807 | 3,075,126,349 | 6,967,589,458 |
| | 2019 | 256,512,476,640 | 365,568,670,314 | 279,457,593,298 | 342,728,188,791 | 22,840,481,523 | 5,435,794,359 | 26,715,497,527 | 6,449,707,777 | 6,834,970,432 |
| | 2020 | 274,829,861,346 | 386,085,725,196 | 290,719,342,438 | 361,598,786,961 | 24,486,938,235 | 8,907,652,128 | 24,291,453,339 | 6,466,977,395 | 6,705,777,925 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2021 | 290,459,816,948 | 410,316,706,779 | 298,372,561,654 | 386,009,303,282 | 24,307,403,497 | 10,740,526,922 | 20,384,114,425 | 6,132,858,523 | 5,906,433,556 |
| | 2022 | 312,068,149,520 | 409,085,360,391 | 316,697,269,309 | 384,659,422,934 | 24,425,937,457 | 8,195,899,671 | 21,455,624,090 | 5,481,854,684 | 5,245,437,808 |
| | 2023 | 317,122,559,016 | 434,279,583,525 | 354,774,379,687 | 409,810,370,743 | 24,469,212,782 | 8,351,036,338 | 25,508,636,581 | 5,011,410,036 | 4,813,574,434 |
| **AL-ARAFAH ISLAMI BANK PLC** | 2013 | 125,715,395,419 | 173,161,626,610 | 140,980,556,620 | 157,070,456,605 | 16,091,170,005 | 5,448,000,000.00 | 17,966,328,508 | 4,948,561,757 | 1,328,205,351 |
| | 2014 | 146,740,366,319 | 206,548,713,972 | 166,851,166,195 | 192,279,482,513 | 14,269,231,459 | 5,000,000,000.00 | 19,725,917,293 | 6,333,450,240 | 1,490,190,743 |
| | 2015 | 162,503,144,128 | 229,106,656,939 | 169,887,081,401 | 209,870,582,418 | 19,236,074,521 | 5,228,000,000.00 | 18,568,005,614 | 6,360,157,867 | 1,615,987,762 |
| | 2016 | 196,519,377,941 | 272,900,048,477 | 199,703,920,008 | 251,562,560,539 | 21,337,487,938 | 5,180,000,000.00 | 18,830,156,209 | 7,511,031,922 | 2,199,578,985 |
| | 2017 | 235,905,234,524 | 319,255,292,509 | 244,806,263,475 | 296,734,617,727 | 22,520,674,782 | 8,324,000,000.00 | 20,491,028,035 | 6,955,769,554 | 2,259,155,807 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2018 | 261,874,133,800 | 338,465,295,212 | 266,205,492,472 | 314,982,065,956 | 23,483,229,256 | 6,528,000,000.00 | 24,956,812,132 | 6,308,895,606 | 1,908,962,937 |
| | 2019 | 288,486,023,705 | 381,051,811,932 | 297,241,809,055 | 356,903,601,299 | 24,148,210,653 | 6,212,000,000.00 | 28,811,650,830 | 7,860,627,076 | 1,531,420,060 |
| | 2020 | 308,620,664,430 | 418,298,652,427 | 326,023,411,262 | 392,842,807,715 | 25,455,844,712 | 8,544,000,000.00 | 26,331,277,005 | 6,687,762,748 | 1,732,137,218 |
| | 2021 | 336,890,715,963 | 462,687,049,884 | 353,287,966,110 | 436,786,707,658 | 25,900,342,226 | 9,800,000,000.00 | 23,446,012,634 | 7,494,140,347 | 1,823,372,676 |
| | 2022 | 406,787,794,869 | 534,756,109,343 | 394,045,837,511 | 508,838,615,037 | 25,917,494,306 | 8,764,000,000.00 | 25,221,415,249 | 8,493,513,447 | 1,841,051,480 |
| | 2023 | 447,121,447,955 | 573,308,834,446 | 419,395,249,636 | 546,811,209,029 | 26,497,625,417 | 9,028,000,000.00 | 31,299,693,298 | 8,234,392,319 | 1,802,540,592 |
| **BANK ASIA PLC** | 2013 | 107,942,564,235 | 165,067,196,178 | 133,682,889,808 | 150,509,777,122 | 14,557,419,056 | 7,610,000,000 | 14,718,904,894 | 5,300,697,553 | 699,449,908 |
| | 2014 | 119,890,004,570 | 184,093,842,368 | 140,681,116,682 | 167,491,743,264 | 16,602,099,104 | 8,500,000,000 | 13,963,856,420 | 5,771,998,282 | 1,229,245,329 |

**Total Loans and Advances:**

The study reveals a consistent upward trend in loans and advances across all four banks, highlighting growth in lending activities. BRAC Bank PLC led the expansion, reaching BDT 520 billion in 2023, reflecting its aggressive financial growth. Meanwhile, Al-Arafah Islami Bank Limited PLC experienced a significant surge in 2016, marking a key turning point in its lending portfolio. The remaining banks, AB Bank PLC and Bank Asia, demonstrated steady and sustained growth over the years, maintaining a stable upward trajectory.

**Total Assets:**

The study has demonstrated a steady upward trend in total assets, with BRAC Bank leading the pack, surpassing BDT 800 billion in 2023, a 16.3% annual growth since 2013. Other banks, including Bank Asia, AB Bank, and Al-Arafah Islami Bank, have maintained asset values within the BDT 400–600 billion range, showing consistent but varied growth. Among them, AB Bank PLC has exhibited the slowest growth, reaching approximately 8% over the same period.

**Total Deposits:**

Among the four banks, BRAC Bank PLC recorded the highest deposit volume, reaching BDT 588 billion in 2023, maintaining a strong upward trajectory over the years. However, a notable anomaly appeared in Bank Asia PLC's deposit trend, as it experienced a sharp decline to BDT 18 billion in 2020, contrasting with the steady and consistent growth observed in the other banks over the past 11 years.

**Total Liabilities:**

The trend in liabilities reflects a steady upward trajectory, mirroring the growth in assets across all banks. BRAC Bank PLC recorded the highest liabilities, exceeding BDT 700 billion in 2023, underscoring its expanding financial commitments. Al-Arafah Islami Bank PLC followed closely, reaching BDT 550 billion, maintaining its steady rise in liabilities. This consistent increase suggests a scaling-up of banking operations, likely driven by rising deposits and lending activities.

**Total Shareholders' Equity:**

Total equity has shown a consistent upward trend, with BRAC Bank PLC leading the growth, nearing BDT 90 billion in 2023. In contrast, Bank Asia, AB Bank, and Al-Arafah Islami Bank maintained equity values ranging between BDT 20 billion to BDT 30 billion, with AB Bank PLC exhibiting the slowest expansion at only 3.2% over the period.

**Market Capitalization:**

In 2023, BRAC Bank PLC achieved an estimated market capitalization of BDT 57 billion, reflecting an impressive 27% growth from 2013. Meanwhile, AB Bank PLC experienced significant volatility, with its market cap hitting a low of BDT 5 billion in 2019 before recovering to a modest 1.4% growth rate overall. This instability suggests fluctuations in investor sentiment and market performance over the years.

**Interest Income:** An Increasing upward trend can be observed in the interest income of the four banks. However, Brac Bank PLC shows the highest growth of 7.8%, as AB Bank PLC faces volatility, and the other banks show a stable growth. Bank Asia also experienced a downturn in interest earnings in 2020.

**EBIT:**

BRAC Bank PLC has the highest EBIT total (BDT 103.6 billion) with a peak of BDT 15.74 billion in 2023 and a consistent upward trend over the years. AB Bank PLC showed more volatility, with a lower total EBIT of BDT 61.7 billion, peaking in 2014 (BDT 6.92 billion)

but also experiencing dips. While Al-Arafah Islami Bank PLC maintained stable earnings, reaching BDT 8.23 billion in 2023, with a steady upward trajectory, Bank Asia PLC exhibited strong growth, with EBIT rising from BDT 5.3 billion in 2013 to BDT 11.71 billion in 2023, showcasing consistent expansion.

**Retained Earnings:**

BRAC Bank PLC has exhibited the strongest retained earnings growth, reaching BDT 33.19 billion in 2023, with an impressive 20.5%. On the other hand, AB Bank PLC has faced declining retained earnings, dropping to BDT 4.81 billion in 2023, marking a -0.7% growth. Al-Arafah Islami Bank PLC maintained moderate stability, with retained earnings increasing at 3.2%, reaching BDT 1.80 billion in 2023. Meanwhile, Bank Asia PLC showed improvement, growing its retained earnings from BDT 699 million in 2013 to BDT 3.35 billion in 2023, achieving a strong 16.9% over the period. This trend shows BRAC Bank and Bank Asia showing significant growth, while AB Bank struggled with retention losses.

For further analysis, two regression models were developed, which are the following:

**Model 1:**

$$\widehat{Z\,Score} = \widehat{\beta_0} + \widehat{\beta_1} ROA + \widehat{\beta_2} ROE + \widehat{\beta_3} CAR + \epsilon$$

Model 2:

$$\widehat{TQ} = \widehat{\beta_0} + \widehat{\beta_1} Z\,Score + \widehat{\beta_2} NPL + \epsilon$$

Where model 1 examines the relationship of Z score (Dependent Variable) on ROA, ROE, and CAR(Independent Variable), and model 2 examines the relationship of Tobin's Q (Dependent Variable) on Z score and NPL (Independent Variable).

## Results and Discussion

### ROA

Return on Assets (ROA) tends to show a downward trend for all the banks. However, Brac Bank PLC peaked the highest among all banks, gaining 1.72% Return on Assets around 2017, where in the same year AB Bank PLC saw a huge dip to 0.1% return on assets. Bank Asia PLC and Al Arafah Islami Bank Limited PLC show the less volatile downward trend of ROE over the period.

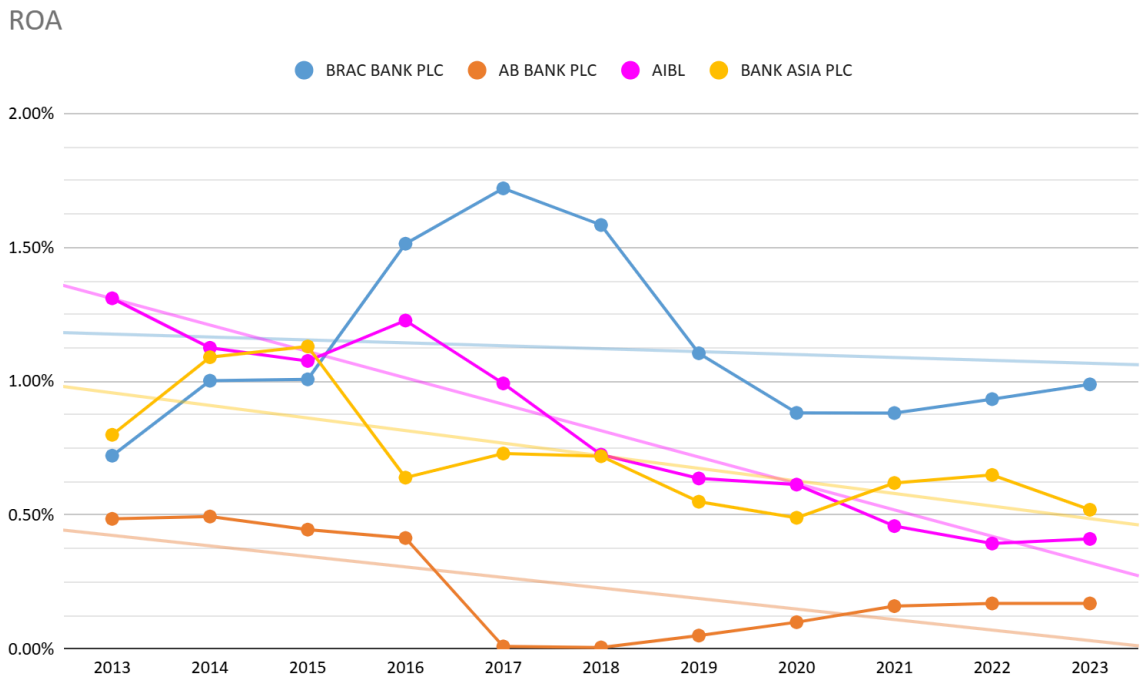



**Fig: ROA (2013-2023)**

### ROE

Return on equity also shows a downward trend, with Brac Bank PLC Peaking at the highest over 20% in 2017, while Al Arafah Islami Bank Limited PLC shows a huge dip to less than 5% Return on equity in 2014.

AB Bank PLC also experienced a dip in 2017 , when the bank earned only 0.13% return on total equity in 2017 and continued to struggle with the downturn in 2018. Bank Asia on the other hand, shows the most stable return out of all banks in the given period.

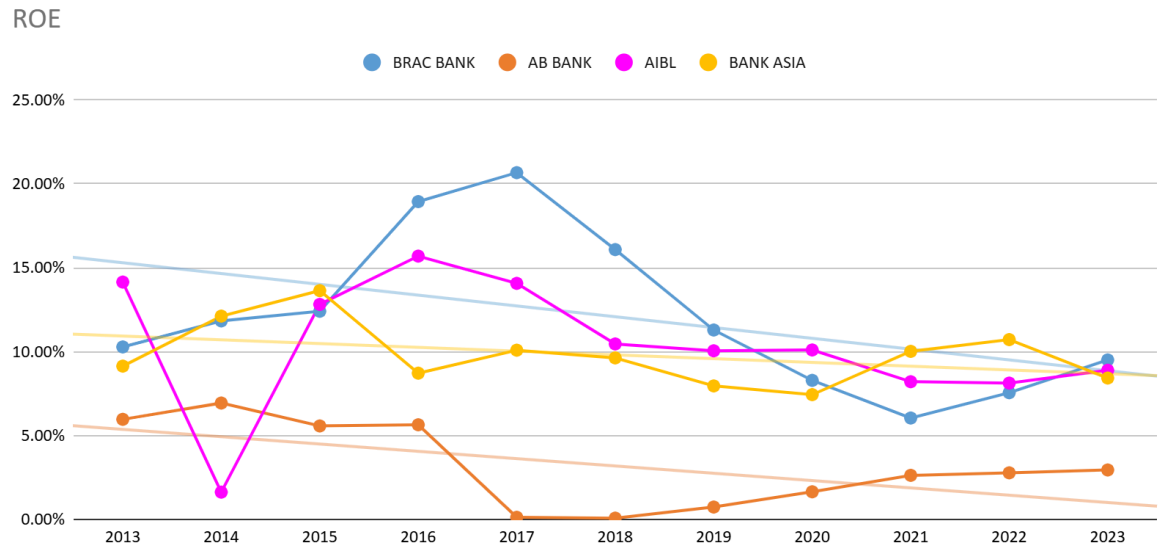



**Fig: ROE (2013-2023)**

**CAR:**

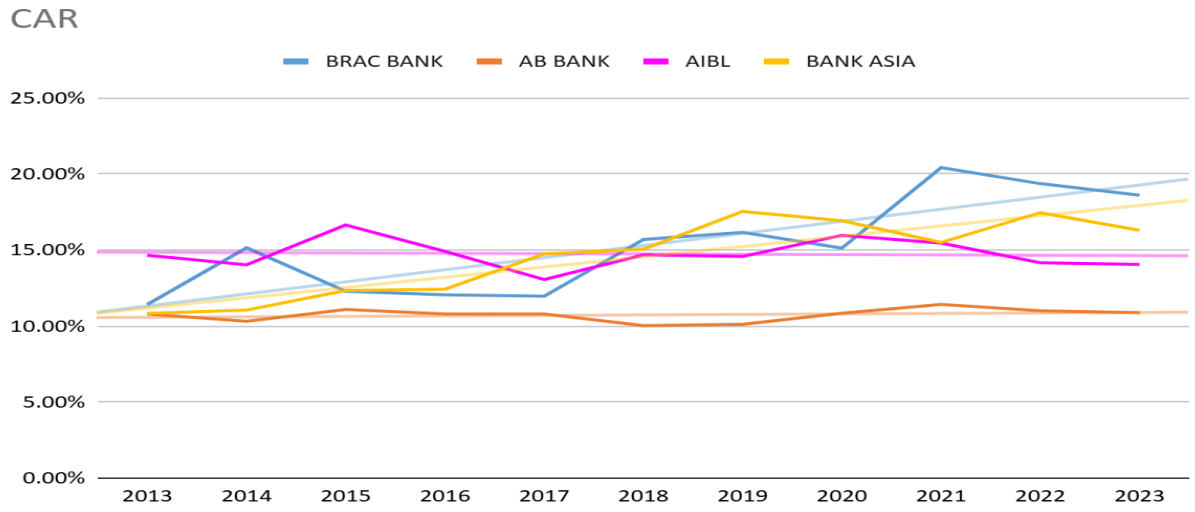


**Fig: CAR(2013-2023)**

Capital Adequacy Ratio shows an upward trend over the period, with Brac Bank PLC reaching 20% in 2021, followed by a stable growth of Bank Asia PLC, reaching over 15% in 2019. Al Arafah Islami Bank Limited PLC also saw an increase in 2015, however, CAR started to decrease in the following year. AB Bank is observed to have the lowest CAR, barely able to meet the minimum requirements of 10% over the period.

**Tobin's Q**

TQ Ratio shows an upward trend, with Brac Bank PLC having the highest of value of 1.06, followed by AIBL, ranging between 1 to 1.05. AB Bank and Bank Asia show stable growth over the period, which shows overvaluation of each bank shares.

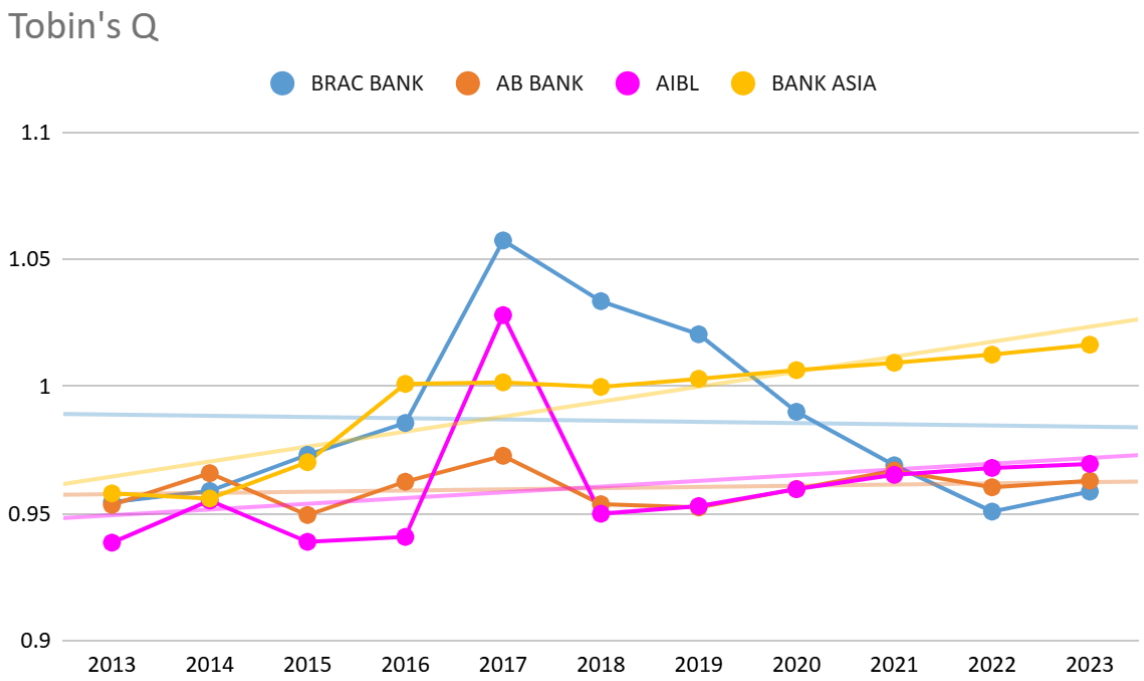



**Fig: Tobin's Q (2013-2023)**

**NPL**

Over the years, Non-Performing Loans show an increasing trend. While AIBL, Brac Bank PLC, and Bank Asia are observed to have similar growth in NPL over the years, AB Bank shows a significant increase of over 30% around the end of the 2010s. After 2019 NPL of AB Bank experienced a downturn, but from 2021, AB Bank PLC again started to show an increasing trajectory of Non-Performing Loans.

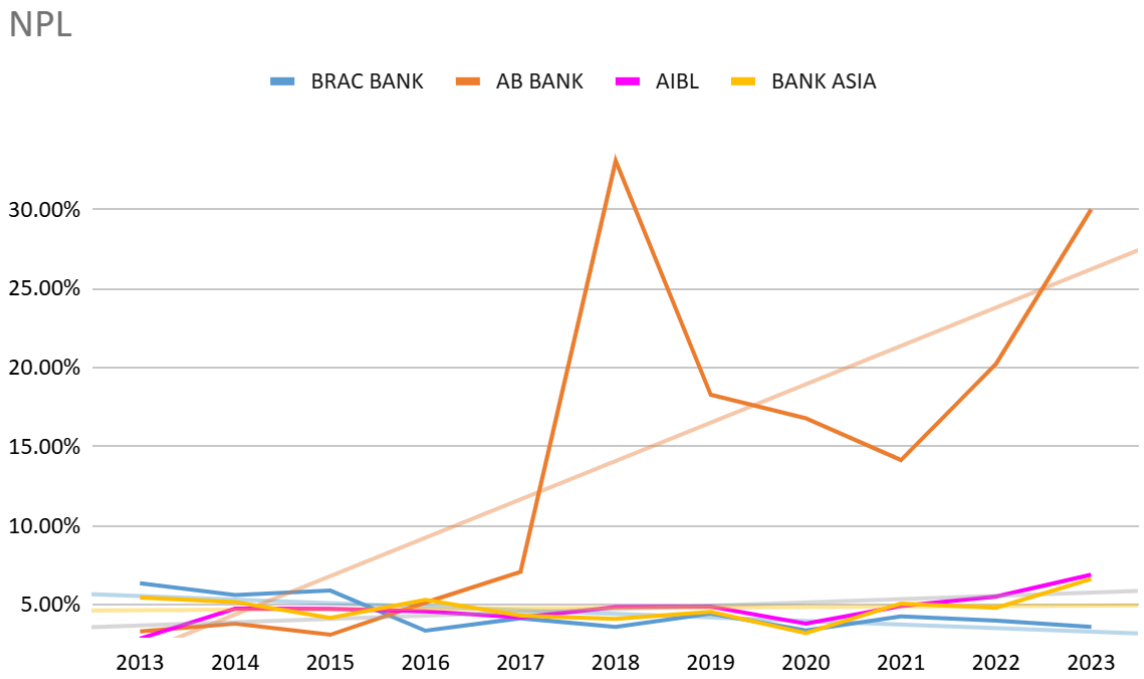



**Fig: NPL (2013-2023)**

## Data analysis and results

***All the analyses were carried out at a 95% level of significance. - the p values are recorded accordingly.***

Null Hypothesis (H0) : $\beta_1 = \beta_2 = \beta_3 = 0$

Alternative Hypothesis (H1):$\beta_1 \neq \beta_2 \neq \beta_3 \neq 0$

**Table 1 - Estimation of results of model 1**

| Dependent variable: Z score; number of observations: 44; R squared: 0.055; Adjusted R |
|---|

| squared: - 0.00152; F significance: 0.50974 | | | | |
|---|---|---|---|---|
| | Coefficients | Standard error | T stat | P value |
| Intercept | -0.67968 | 1.64045 | -0.41425 | 0.68090 |
| ROA | -106.971 | 148.766 | -0.71905 | 0.47629 |
| ROE | 11.8358 | 12.8831 | 0.91871 | 0.36375 |
| CAR | 13.9698 | 12.1872 | 1.14626 | 0.25849 |

As per the estimation presented in table 1, we failed to reject our null hypothesis. None of the explanatory variables proved to be statistically significant. And the coefficient of ROA, had the incorrect sign. The expected effect of ROA on Z score was positive, meaning as the bank's ROA increases the Z Score also increases. This signifies a fall in the probability of default. Alternatively, the estimated result could hypothetically mean that, despite higher ROA which portrays more efficient use of resources by banks, given that there were large fluctuations influenced by short term changes net income and assets within the sample banks and the inability of ROA to prioritise assets age and condition, might have made long term predictions more uncertain which is true in the case of Bangladesh. Therefore heightening the possibility of default risk at particular point in time when the study was conducted. Economically such a relationship does not exist. To interpret the results, as per table 1 an increase in ROA reduces Z score by 106.97 points which may not be entirely realistic as the Z score ranges between 0 and 4. However, in the case of ROE and CAR, the expected effect on Z score were realised. The positive effect of both ROE and CAR on Z score is supposedly positive. A rise in both ROE portrays that a bank's ability to generate a return as a ratio of its equity has increased implying boasted owners and investors confidence on the financial performance of the bank ultimately lowering the probability of default. Similarly, in the case of CAR, a positive relationship indicates that a bank has resources readily available to cover or to absorb losses without the bank having to sell off its assets at a low price. Moreover, the

overall significance of the model as captured by the F test also proved to be insignificant. To interpret the results given in table 1, a rise in both ROE and CAR by 1% each, results in a rise in the Z score by 11.8358 and 13.9698 points respectively. These variations of results may be more realistic given the nature of the dependent variable.

**Table 2 - Estimation results of model 2:**

Null Hypothesis (H0) : $\beta_1 = \beta_2 = 0$

Alternative Hypothesis (H1):$\beta_1 \neq \beta_2 \neq 0$

| Dependent variable: Tobin's Q (TQ); number of observations: 44; R Squared: 0.17113; Adjusted R Squared: 0.13070; F significance: 0.0213 | | | | |
|---|---|---|---|---|
| | Coefficients | Standard error | T stat | P value |
| Intercept | 0.9717 | 0.00711 | 136.660 | 3.75458E-56 |
| Z score | 0.00501 | 0.00204 | 2.46135 | 0.001814 |
| NPL | -0.05928 | 0.06198 | -0.95634 | 0.344505 |

In the case of model 2, as per the estimation presented in table 2 we succeeded in rejecting the null hypothesis in almost all levels. Even though the overall significance of the model is below the 5% threshold, empirically we could not prove the statistical significance of NPL to Tobin's Q in comparison to Z score. However, model 2 succeeds in establishing the proper economic relationships between the variables. In the case of NPL, it is widely accepted that a higher NPL negatively affects a banks' market value. A high NPL can reduce a bank's earnings and overall value that investors and other stakeholders place on the bank. Therefore, stakeholders perceive lower levels of profitability in the future which in turn affects banks TQ ratio. For the interpretation of the results, an increase in the NPL by 1% would result in a fall in Tobin's Q by 0.059 units (5.9% expressed in percentages). Z scores on the other hand have a distinct application from that of Tobin's Q for even though both of these variables report on a bank's financial health, they focus on different aspects. However, we postulate

that a higher Z score can have an effect on the Tobin's Q of a bank establishing a positive relationship between the variables. Given Z score assesses that probability of default, a higher probability would indicate a loss of trust among potential and existing stakeholder groups. Therefore, the effects can negatively spillover on the bank's Tobin's Q which assesses the market value of a bank. Hence, we believe that a higher Z score would naturally lead to a higher Tobin's Q. As per the results, an increase in Z score by 1% would increase Tobin's Q by 0.00501 units (0.5% expressed in percentages).

## Conclusion

The above study examined the impact of the mentioned key financial ratios, which include ROA, ROE, CAR, NPL and Z-score, in order to examine the financial stability and market valuation of banks in Bangladesh using two separate models. The first model examined the effects of ROA, ROE and CAR on the Z score, which is a widely accepted measure to calculate a bank's default risk. The later model tried to analyze the relationship between NPL, Z-score and Tobin's Q- which represented market perception of a bank's value. In the case of Model 1, the results have shown that none of the mentioned explanatory variables had statistically significant effects on the Z-score at a 95% confidence level. The positive relationships between ROE and CAR with Z-score can be observed, but the coefficient of ROA unexpectedly showed a negative sign. This finding contradicts with the theoretical assumption that higher ROA-indicator of better resource utilization-should lead to a higher Z-score, thereby reducing the probability of default. This consistency could stem from data limitations, including short-term fluctuations in net income and total assets within the observed banks, or the inability of ROA to reflect the age and the quality of those assets. Economically, a negative ROA-Z-score relationship does not hold strong ground, especially given the unrealistic magnitude of effect implied by the coefficient. In contrast, Model 2

yielded more empirically sound and statistically significant results. The model showed a significant relationship between the explanatory variables and Tobin's Q, with a relatively higher R-squared value. Notably, an increase in NPL was found to negatively impact Tobin's Q, which aligns with economic theory—higher non-performing loans reduce bank earnings and stakeholder confidence, thereby reducing market valuation. The Z-score also had a positive and economically intuitive effect on Tobin's Q, suggesting that a more financially sound bank, with a lower risk of default, is perceived more favorably by the market. While the results of Model 2 support the hypothesized relationships and offer valuable insights into the determinants of bank valuation, several limitations must be acknowledged. Firstly, the dataset exhibited characteristics of non-stationary time series data, limiting the reliability of long-term predictions. Secondly, the models did not account for potential structural breaks or inter-bank heterogeneity. Employing panel data techniques such as Fixed Effects, Random Effects, or the Least Squares Dummy Variable (LSDV) model could have provided more robust estimations by capturing bank-specific variations. Moreover, the unexpected sign of ROA in Model 1 may be indicative of data or specification issues, further underlining the need for enhanced model diagnostics. Tests such as the Lagrange Multiplier (LM) test were not conducted and could have helped identify autocorrelation or specification bias within the models. In conclusion, while Model 2 demonstrates a stronger empirical and economic foundation in assessing market valuation through Tobin's Q, Model 1 suffers from both statistical insignificance and potential misspecification. The study highlights the importance of financial health indicators in influencing investor perception, but it also underlines the challenges of using static linear models on complex financial datasets. Future research should incorporate dynamic panel models, more extensive time series data, and diagnostic testing to yield more robust insights into the financial stability and valuation of banks in developing economies like Bangladesh. For policymakers, regulators, and bank management in

developing economies such as Bangladesh, the findings suggest that maintaining sound equity returns, adequate capital reserves, and low levels of NPLs are critical not only for ensuring institutional solvency but also for enhancing investor confidence and market standing.

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